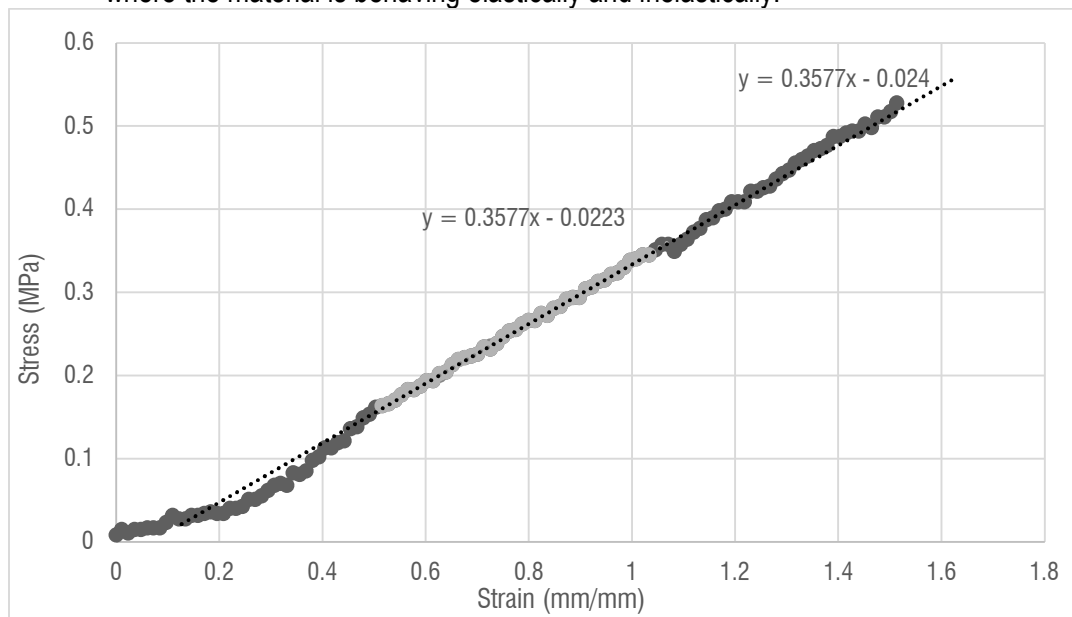


MILESTONE 3 (STAGE 1) – MATERIALS TESTING LAB

Team Number: 15

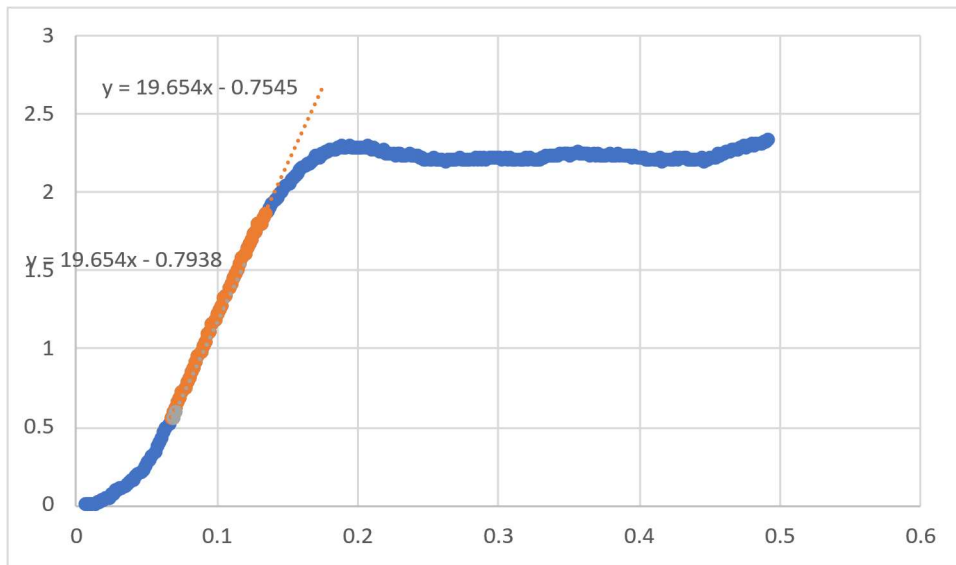
Summary of Submitted Work

- Plot a graph of your stress vs. strain curve for the **Tensile Test**. Indicate the regions on the curve where the material is behaving elastically and inelastically.



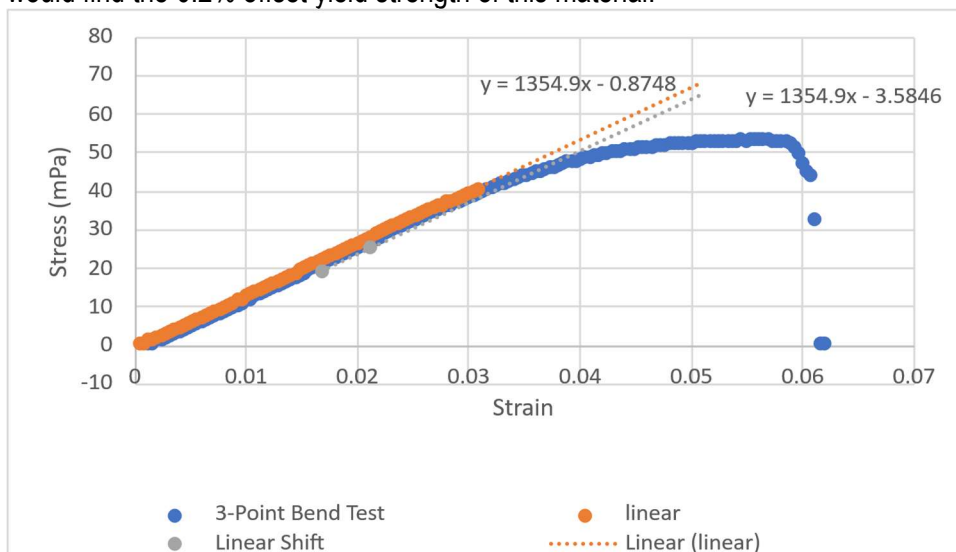
The material behaves elastically when the stress-strain relationship is linear, indicated by the light grey region of the graph, and inelastically when the stress-strain relationship is not linear, indicated by the black region of the graph.

- Plot a graph of your stress vs. strain curve for the **Compression Test**. Briefly describe how you would find the elastic modulus from this curve.



The elastic modulus can be determined by regarding the section of the graph where the stress-strain relationship is linear and determining the slope of this region with a line of best fit. In this case the elastic modulus is 19.654 mPa.

3. Plot a graph of your stress vs. strain curve for the **Three-Point Bending Test**. Briefly describe how you would find the 0.2% offset yield strength of this material.



To find the 0.2% offset yield strength, a line of best fit should be fitted on the linear region of the graph, and the line should be shifted by 0.002, or 0.2% to the right.

*** Plots should be created in Microsoft Excel.

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Reflective Questions (answers can be in point form)

1. Why is it ideal to perform tensile, compressive, and three-point bend tests when they all provide stress, strain, and modulus? Is it necessary to perform all three for a single material?

- Real biomaterials, especially implants, will experience tensile, compressive, and 3-point bend forces, all of which have different relationships with stress and strain
- Each test determines how a material responds to different forces: a tensile test for tension, a compressive test for compression, and a 3-point bend for both forces.
- Although they all provide stress, strain and modulus, these values differ across tests, suggesting that materials may have different properties in response to different stresses
- Materials are also usually anisotropic, which means the directionality of force will affect how the material changes in result; doing 3 tests better accounts for an anisotropic materials properties

2. What are some limitations of these tests, individually and as a whole, with respect to implant design decisions? How often are uniaxial loads placed on our joints?

- The tensile and compressive tests both only test forces of one kind, and they both assume a uniaxial load
- The 3-point bend test, tensile test, and compressive test may all yield different results if the material is anisotropic.
- Uniaxial loads on joints are not realistic, generally many forces are present in joints (eg. hip joints experience a compressive force and a bending force).
- So a general limitation is the fact that the tests are conducted in an isolated system, since all the tests are conducted individually and don't mimic all the forces that a joint or bone would feel.